Ruby lattice: ED geometry and parameter conventions

Spinless t-V model | 6 orbitals / primitive unit cell | PBC clusters

1. Model and hopping sign

H0 = sum_[R,(i,j,dr)] t_ij c^dag_(R,i) c_(R+dr,j) + h.c.

Hint = V sum_R [n0 n1 + n0 n2 + n1 n2 + n3 n4 + n3 n5 + n4 n5]

t = +1.0 (triangle), t1 = +0.5, t2 = +0.5 (inter-triangle).

Only the six red triangle edges carry V; no inter-triangle density V.

The +t convention follows the earlier code; do NOT replace it by -t.

2. Primitive cell numbering

Triangle A: orbitals 0, 1, 2. Triangle B: orbitals 3, 4, 5.

Primitive vectors (unit-length bonds):

a1 = (-(1+sqrt(3))/2, -(3+sqrt(3))/2)

a2 = (-(1+sqrt(3))/2, (3+sqrt(3))/2)

0: (0, sqrt(3)/2) 3: (0, -1-sqrt(3)/2)

1: (-1/2, 0) 4: (-1/2, -1)

2: (1/2, 0) 5: (1/2, -1)

3. Independent primitive hopping representatives

A bond (i,j,dx,dy) means (R,i) -> (R+(dx,dy),j), plus h.c.

Class	(i,j,dx,dy)			amplitude
t	(0,1, 0, 0)	(0,2, 0, 0)	(2,1, 0, 0)	1.0
t	(3,4, 0, 0)	(3,5, 0, 0)	(4,5, 0, 0)	1.0
t1	(1,4, 0, 0)	(5,0, 0,-1)	(2,3,-1, 0)	0.5
t2	(3,1, 0,-1)	(2,5, 0, 0)	(0,4,-1, 0)	0.5

4. PBC site ordering and verification

A site is indexed by s = 6 * cell_index + orbital (0-based).

Store one directed representative per undirected bond; add h.c.

The bond table contains canonical and plotted-image winding separately.

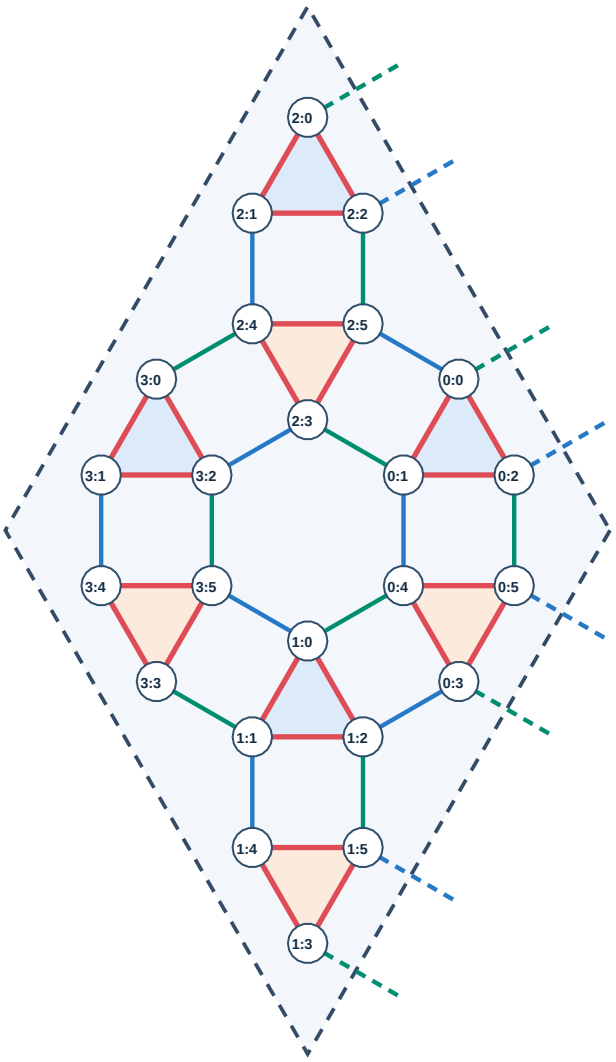
Every site has coordination four. All three clusters pass the folded-band check.

Source convention: ruby_selection_rule_check_physical_labels.py

Full link data: ed_benchmark/pbc_bond_list.csv

2 x 2 PBC

Sites 24 | independent bonds 48 | displayed PBC wraps 8 | z = 4



Cross-boundary PBC links

src -> dst (display winding)

- 0:5 -> 2:0 (0,-1)
  t1; canonical (0,-1)
- 0:2 -> 1:3 (-1,0)
  t1; canonical (-1,0)
- 0:3 -> 2:1 (0,-1)
  t2; canonical (0,-1)
- 0:0 -> 1:4 (-1,0)
  t2; canonical (-1,0)
- 1:5 -> 3:0 (0,-1)
  t1; canonical (0,-1)
- 1:3 -> 3:1 (0,-1)
  t2; canonical (0,-1)
- 2:2 -> 3:3 (-1,0)
  t1; canonical (-1,0)
- 2:0 -> 3:4 (-1,0)
  t2; canonical (-1,0)

Red: t=1; blue: t1=0.5; green: t2=0.5.

Solid = local bond. Dashed = bond to periodic image (h.c. implicit).

All PBC wrap links are retained; none is dropped due to a negative R.

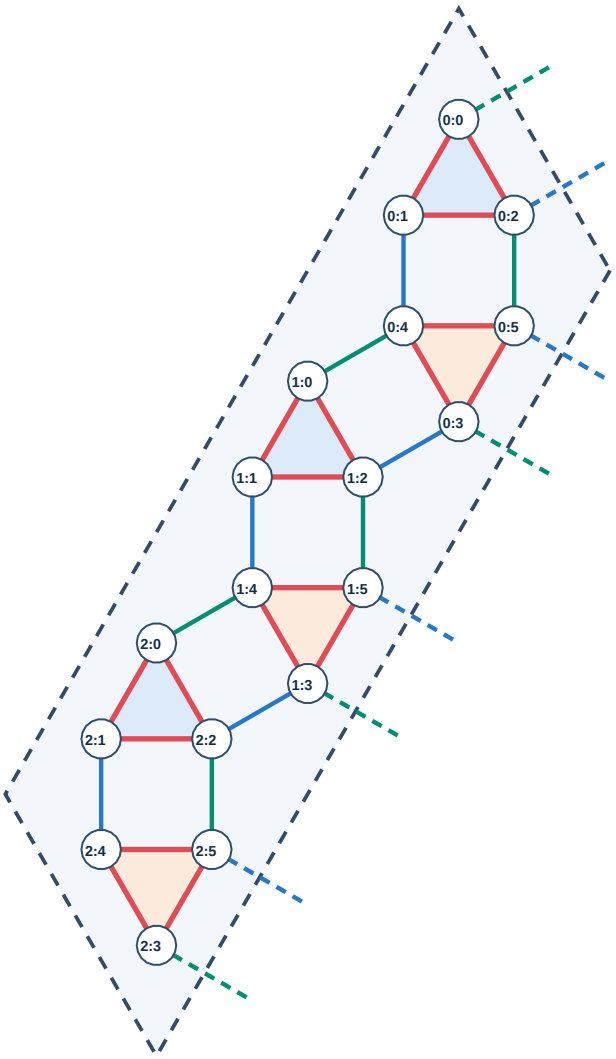
Supercell basis (columns in units of primitive a1,a2):

A1 = 2 a1 + 0 a2; A2 = 0 a1 + 2 a2

Use pbc_bond_list.csv for ED; plotted images do not change site IDs.

3 x 1 PBC

Sites 18 | independent bonds 36 | displayed PBC wraps 8 | z = 4



Cross-boundary PBC links

src -> dst (display winding)

- 0:5 -> 0:0 (0,-1)
  t1; canonical (0,-1)
- 0:2 -> 2:3 (-1,0)
  t1; canonical (-1,0)
- 0:3 -> 0:1 (0,-1)
  t2; canonical (0,-1)
- 0:0 -> 2:4 (-1,0)
  t2; canonical (-1,0)
- 1:5 -> 1:0 (0,-1)
  t1; canonical (0,-1)
- 1:3 -> 1:1 (0,-1)
  t2; canonical (0,-1)
- 2:5 -> 2:0 (0,-1)
  t1; canonical (0,-1)
- 2:3 -> 2:1 (0,-1)
  t2; canonical (0,-1)

Red: t=1; blue: t1=0.5; green: t2=0.5.

Solid = local bond. Dashed = bond to periodic image (h.c. implicit).

All PBC wrap links are retained; none is dropped due to a negative R.

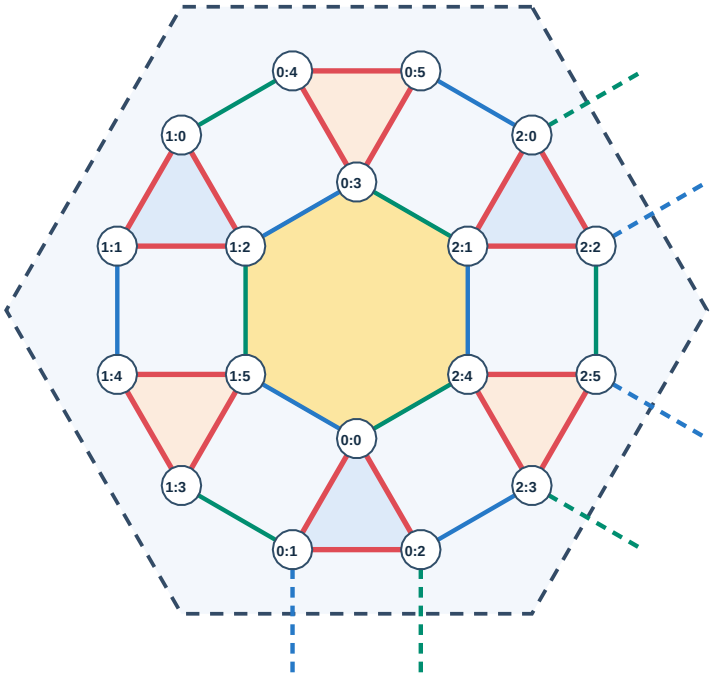
Supercell basis (columns in units of primitive a1,a2):

A1 = 3 a1 + 0 a2; A2 = 0 a1 + 1 a2

Use pbc_bond_list.csv for ED; plotted images do not change site IDs.

Six-triangle ring PBC (area 3)

Sites 18 | independent bonds 36 | displayed PBC wraps 6 | z = 4



Cross-boundary PBC links

src -> dst (display winding)

- 0:1 -> 0:4 (1,0)
  t1; canonical (0,0)
- 0:2 -> 0:5 (1,0)
  t2; canonical (0,0)
- 2:5 -> 1:0 (0,-1)
  t1; canonical (0,0)
- 2:2 -> 1:3 (-1,-1)
  t1; canonical (-1,0)
- 2:3 -> 1:1 (0,-1)
  t2; canonical (0,0)
- 2:0 -> 1:4 (-1,-1)
  t2; canonical (-1,0)

Red: t=1; blue: t1=0.5; green: t2=0.5.

Solid = local bond. Dashed = bond to periodic image (h.c. implicit).

Six whole triangles surround a physical hexagon with alternating t1/t2.

Supercell basis (columns in units of primitive a1,a2):

A1 = 1 a1 + -1 a2; A2 = 1 a1 + 2 a2

Use pbc_bond_list.csv for ED; plotted images do not change site IDs.